
TITLE PLACEHOLDER: TIME COMPRESSION ALLOWS LEARNING OF EMERGENT STRUCTURES *

ABSTRACT

Keywords Artificial life · Representation learning · Emergence · Simulations

1 Introduction

In science, a wide variety of problems require running simulations for obtaining the final results. For example, in physics, the *no fast-forwarding theorem* states that a quantum system can't be simulated faster than its evolution according to time. However, if we only care about some macroscopic final properties, can we fast-forward or even skip simulation? This is similar to humans' System 1 thinking — e.g., when we drive cars, we humans don't run step-by-step simulations in our brains before we make decisions on next moves, but rather, we run high-level simulations in the abstraction level, which allows us to predict from step t to $t + \Delta t$, where Δt might be large. Such macroscopic simulations correspond to "intuitive" physics, which is imprecise but retains key information we want — this is in contrast to "real" physics that we employ in e.g. physical experiments.

This motivates the hypothesis that a good abstraction correlates with the capability to predicting macroscopic properties ahead of time. Specifically, a good abstraction might enable such capability, and/or predicting macroscopic properties ahead of time gives rise to a good abstraction. In this paper, we test and demonstrate the latter.

As a testbed, we picked an artificial life (ALife) substrate, Game of Life. The problem is: Can AI models trained on Game of Life learn high-level representations of "emergent structures" in Game of Life? Game of Life rules are easy to learn as they just involve convolution, but to predict several steps ahead is a much harder task, as the system is very chaotic and a tiny difference in initial state can result in very different states in several steps. We show that models asked to predict more steps ahead contain better, i.e., more linearly separable, representation spaces.

2 Methodology

2.1 Task Selection

Game of Life (GoL) has clear deterministic rules, and in particular, Conway's GoL is Turing-complete. Predicting the immediate next state ($t \rightarrow t + 1$) is an easy task, but predicting multiple steps ahead ($t \rightarrow t + \Delta t$, where $\Delta t > 1$) is a much harder task. Thus, we expect that a model has to make more mistakes and compress information more efficiently as Δt increases. To test on a variety of simulations not limited to Conway's GoL, we also used other GoL rules found by the ASAL framework by [1].

2.2 Model

We train ConvNet that maps from GoL state t to state $t + \Delta t$ in one forward pass. Specifically, the ConvNet takes a 2D GoL grid as input, encodes it, and uses a 1×1 convolution to put it into a D -dimensional feature map. It adds a learned embedding for Δt , the number of timesteps skipped ahead, to each spatial location. Then, it applies a series of residual blocks. Finally, a 1×1 convolution projects back to the output channels.

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2.3 Representation Space Evaluation

To measure whether the ConvNet’s representations capture the emergent structures in GoL, we use a linear probe from the representations of input GoL frame to CLIP vector encoding the frame. CLIP encoding could be seen as the average of lots of humans’ representations, given its training data, so quantifying how well the representations linearly map to CLIP vectors is more objective than evaluation metrics such as how well the representations encode the number of gliders.

3 Results

Preliminary results show that models asked to skip more timesteps ahead (higher Δt) have better representations of input states (states t), which is reflected as linear probe having higher R^2 to CLIP vector (as shown in the middle panel of Figure 1). The representations of target states (states $t + \Delta t$) don’t necessarily improve as Δt increases (as shown in the right panel of Figure 1). But when compared against the baseline of linear probing a random-weights model, the representations of target states do surpass the baseline’s as Δt increases.

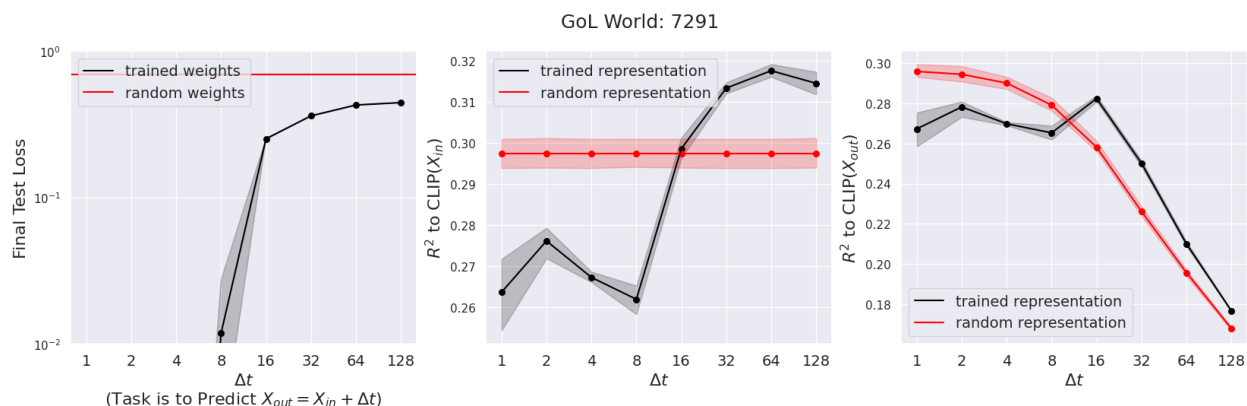


Figure 1: Results on Game of Life 7291. Left: ConvNet training result. Middle: Linear probe result for input state. Right: Linear probe result for target state.

To analyze the quality of representations, we also visualize models’ predictions. For example, Figure 2 below shows $\Delta t = 16$ model’s performance on GoL 7291. When comparing the model’s prediction in the third row to the ground-truth target states in the second row, we can see that the model is learning to capture abstract patterns more confidently, without caring too much about predicting each cell precisely in the more chaotic regions. As an analogy, the model captures the "organisms" in GoL, without caring too much about what molecules or atoms are present. This is exactly what we hope to see from a model that learns to capture the "emergent structures" of GoL.

4 Discussion

What is so special about compressing time? How is it different from previous approaches that compress space and get better representations? I.e., what new things does time bring us versus if we just treat the time dimension as an additional space dimension?

Things we can try next? Thinking about some downstream tasks, as good representations should enable us to do more things, such as:

- Build a taxonomy of emergent patterns (like the periodic table idea Kaiming mentioned once)
- Find rare patterns (model right now perhaps only allows us to find patterns that reach a certain final state in Δt steps, but if we want to find e.g. oscillators that show periodic behavior, then we need model that can be conditioned on any Δt)

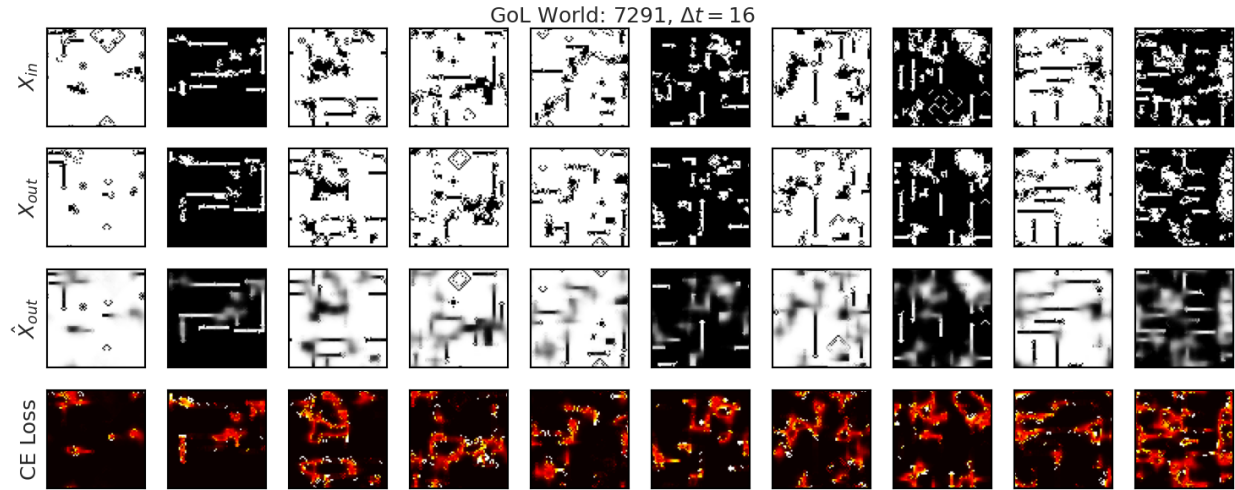


Figure 2: Model performance on Game of Life 7291, with $\Delta t = 16$. First row: Input state. Second row: Ground-truth target state. Third row: Model prediction of target state. Fourth row: Cross-entropy loss.

References

- [1] Akarsh Kumar, Chris Lu, Louis Kirsch, Yujin Tang, Kenneth O. Stanley, Phillip Isola, and David Ha. Automating the Search for Artificial Life with Foundation Models, December 2024.